

1 Latent Representation Probing for Toxic Comment Detection

1.1 Data Card

1.1.1 Dataset Overview

The project is designed for public toxicity benchmarks such as Civil Comments, the Jigsaw Toxic Comment Classification dataset, or an equivalent dataset available through Hugging Face. The target task is binary classification:

- **0 — Non-toxic / benign:** comments that do not meet the project’s toxicity definition.
- **1 — Toxic:** comments containing toxicity such as threats, insults, identity attacks, harassment, or explicit abusive language, as defined by the selected benchmark.

The exact dataset name, release version, licensing terms, and sample counts should be recorded from the final notebook configuration.

1.1.2 Data Composition and Distribution

The recommended evaluation design uses a training split and a held-out evaluation split, with a separate validation split when model selection or threshold tuning requires it. The supplied outline gives an example of an 80% training and 20% test split; this value should be replaced if the notebook uses a different partition. The records consist primarily of short- to medium-length online comments, forum posts, and public discussions. Toxicity datasets are often imbalanced: benign comments may be more numerous, while toxic examples cover a range of severity and category.

Before training, report the number of examples in each split and the class counts for both labels. Do not infer balance from the task definition alone.

1.1.3 Feature Extraction and Processing

A frozen transformer language model—such as a Gemma-family model—can be used to extract hidden states through Hugging Face Transformers or TransformerLens. Intermediate representations may capture semantic and contextual information without updating the base model weights.

A typical preprocessing pipeline includes tokenization, sequence padding or truncation, activation extraction using mean pooling or a last-token representation, and feature standardization before fitting the probe classifier. The precise pooling rule, maximum sequence length, selected layers, and handling of missing or malformed text should be documented in the implementation record.

1.1.4 Known Limitations and Biases

Annotation bias. Annotators may disagree about the boundary between mild toxicity, profanity, satire, and civil disagreement.

Domain limitations. Performance may vary on non-English text, platform-specific slang, code-switching, leetspeak, emerging terms, and communities that are underrepresented in the source data.

Representation limitations. A probe may exploit correlations present in the base model’s activations without fully representing the intended concept of toxicity. Layer choice and pooling strategy can materially affect results.

1.1.5 Recommended Reporting

Publish dataset provenance, license information, split construction, class counts, preprocessing choices, layer and pooling selections, metric definitions, threshold policy, and representative error analyses. Where possible, evaluate subgroup and out-of-distribution behavior rather than reporting a single aggregate score.

Prepared from the supplied project outline; exact counts and benchmark identifiers remain notebook-dependent.